Airbnb Melbourne Data Engineering and Market Analysis Report

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1. Executive Summary

This project analyses the Melbourne, Victoria, Australia Airbnb market using public data from Inside Airbnb.

The objective was to build a reproducible data engineering and analytics workflow that transforms raw Airbnb files into cleaned, profiled, and analytics-ready outputs. The project includes dataset familiarisation, data profiling, cleaning, an enriched listing master table, an analytical DuckDB database, SQL analysis, exploratory data analysis, and statistical testing.

A key dataset limitation was identified during profiling: the Melbourne Inside Airbnb files used in this project did not contain usable price values. The price field in the detailed listings file, the summary listings file, and the calendar file contained no non-null values. For this reason, the final analysis focuses on supply, availability, host concentration, review behaviour, neighbourhood patterns, and quality signals rather than price or revenue.

Key findings include:

* Entire homes/apartments dominate Melbourne Airbnb supply, with 17,703 listings.

* Private rooms represent the second-largest room type, with 6,563 listings.

* The Melbourne neighbourhood has the highest listing count, with 7,832 listings.

* Darebin has the highest estimated occupancy-rate proxy, at 0.7066.

* Flexistayz is the largest host by listing count, managing 292 listings.

* Superhost status is statistically associated with different review score distributions.

* Availability patterns differ across room types, neighbourhoods, and weekend/weekday calendar records.

The project intentionally focuses on one city to prioritise depth, code quality, reproducibility, and clear business interpretation.

2. Objectives and Scope

2.1 Objectives

The main objectives of this project were to:

* Understand the Inside Airbnb Melbourne dataset.

* Profile raw data quality.

* Clean and standardise key fields.

* Create an enriched listing-level analytical dataset.

* Build a simple analytical database model.

* Run SQL-based business analysis.

* Generate EDA outputs and visualisations.

* Apply statistical testing to support analytical findings.

* Document assumptions, limitations, and technical decisions clearly.

2.2 Scope

This project focuses on one city:

Melbourne, Victoria, Australia

A single-city scope was selected because the assignment values depth and quality over attempting every possible task superficially.

2.3 Out of Scope

The following items were intentionally not completed:

* Multi-city comparison

* Machine learning price prediction

* LLM-based review analysis

* Cloud deployment

* Dashboard deployment

* Docker containerisation

These items were deprioritised to focus on core data engineering, reproducibility, EDA, statistical analysis, and clear documentation.

3. Dataset Overview

3.1 Dataset Source

The dataset was sourced from Inside Airbnb.

Files used:

* listings.csv.gz

* listings.csv

* calendar.csv.gz

* reviews.csv.gz

* neighbourhoods.csv

* neighbourhoods.geojson

3.2 Main Entities

The main entities in the dataset are:

* Listings: Airbnb properties available in the Melbourne market.

* Hosts: People or organisations managing one or more listings.

* Calendar records: Daily availability records for each listing.

* Reviews: Guest review records linked to listings.

* Neighbourhoods: Geographic groupings used for location-based analysis.

3.3 Key Relationships

The main dataset relationships are:

```
listings.id = calendar.listing_id
listings.id = reviews.listing_id
listings.neighbourhood_cleansed = neighbourhoods.neighbourhood
listings.host_id identifies hosts
```

3.4 Dataset Limitation: Missing Price Values

During profiling, the Melbourne dataset was found to have no usable price data in the relevant files:

* listings.csv.gz price values were fully missing.

* listings.csv price values were fully missing.

* calendar.csv.gz price and adjusted price values were fully missing.

Because of this limitation, price and revenue analysis were excluded from the final analytical findings. This decision avoids producing unsupported or misleading conclusions.

The final analysis therefore focuses on:

* Listing supply

* Annual availability

* Calendar availability

* Review behaviour

* Review scores

* Host concentration

* Neighbourhood-level supply patterns

* Superhost quality signals

4. Methodology

The project followed a structured data engineering workflow:

1. Raw data validation

2. Data profiling

3. Cleaning and standardisation

4. Feature derivation

5. Dataset enrichment

6. Analytical database modelling

7. SQL analysis

8. Exploratory data analysis

9. Statistical testing

10. Documentation and reporting

The full pipeline can be executed with:

```
python src/run_pipeline.py
```

5. Engineering Approach

5.1 Pipeline Design

The project uses a repeatable Python pipeline controlled by:

```
python src/run_pipeline.py
```

The pipeline runs the following scripts:

1. check_data.py
2. profile_data.py
3. clean_data.py
4. build_master_table.py
5. build_database.py
6. run_sql_analysis.py
7. create_eda_outputs.py
8. create_statistical_analysis.py

5.2 Data Cleaning

Cleaning steps included:

* Parsing date fields.
* Converting percentage fields into decimal format.
* Standardising text fields.
* Validating latitude and longitude values.
* Creating host tenure fields.
* Creating review frequency fields.
* Creating calendar-based weekend and weekday indicators.
* Preserving missing price values rather than imputing unsupported prices.

5.3 Enriched Listing Master Table

The enriched listing_master table combines listing attributes with calendar and review summaries.

Derived fields include:

* available_days
* unavailable_days
* occupancy_rate_estimate
* review_frequency_per_month
* listing_count_neighbourhood
* average_rating_neighbourhood

The occupancy_rate_estimate field is treated as a proxy because unavailable calendar days may represent bookings, blocked dates, maintenance periods, or host-disabled dates.

5.4 Analytical Database

DuckDB was used as the analytical database.

Created tables include:

* listing_master
* clean_calendar
* clean_reviews
* dim_host
* dim_neighbourhood
* dim_room_type
* dim_date
* fact_listing
* fact_calendar_daily
* fact_reviews

This provides a simple analytical star-schema style model for querying listing, calendar, and review behaviour.

6. Data Quality Findings

A data profiling report was generated at:

```
reports/data_quality_report.csv
```

The raw dataset sizes were:

* Listings: 24,491 rows
* Calendar: 8,939,215 rows
* Reviews: 943,744 rows
* Neighbourhoods: 30 rows

The most important data quality finding was that price fields were not usable. The detailed listings file, summary listings file, and calendar file all had missing price values. This directly affected the analytical scope and required the project to focus on non-price market intelligence.

This is a material limitation, but documenting it clearly strengthens the reliability of the analysis. The project avoids unsupported price or revenue conclusions and instead analyses fields with sufficient data quality.

7. SQL Analysis Findings

SQL analysis outputs were generated in:

```
reports/sql_outputs/
```

7.1 Listing Supply by Room Type

The largest room type category was Entire Home/Apt, with 17,703 listings. This category had an average annual availability of 151.37 days and an average of 46.39 reviews per listing.

Private rooms were the second-largest category, with 6,563 listings, average annual availability of 139.74 days, and an average of 18.45 reviews per listing.

This indicates that the Melbourne Airbnb market is strongly shaped by entire-home/apartment supply rather than shared accommodation.

7.2 Neighbourhood Supply

The neighbourhood with the highest listing count was Melbourne, with 7,832 listings. Its average annual availability was 140.12 days and the average number of reviews was 49.14.

This suggests that neighbourhood-level supply concentration is important for market monitoring and operational planning.

7.3 Estimated Occupancy Proxy

The neighbourhood with the highest estimated occupancy-rate proxy was Darebin, based on 584 listings. Its average occupancy-rate estimate was 0.7066, while average annual availability was 106.91 days.

This should be treated as a proxy because calendar unavailability may represent bookings, blocked dates, or host-disabled dates.

7.4 Host Concentration

The largest host by listing count was Flexistayz, managing 292 listings. This host had average annual availability of 94.66 days and average reviews per listing of 5.80.

This suggests that host concentration is an important supply-side feature to monitor, especially when distinguishing casual hosts from professional or high-volume operators.

8. Exploratory Data Analysis Findings

EDA outputs were generated in:

```
reports/eda_outputs/
reports/figures/
```

Generated figures include:

* listing_supply_by_room_type.png

* availability_by_room_type.png

* top_neighbourhoods_by_listing_count.png

* review_score_distribution.png

* reviews_by_room_type.png

* monthly_availability_rate.png

* top_hosts_by_listing_count.png

8.1 Listing Supply by Room Type

Figure: reports/figures/listing_supply_by_room_type.png

Business interpretation:

Entire homes/apartments dominate Melbourne Airbnb supply. This suggests the platform is not only serving shared accommodation use cases, but also a large whole-property short-term rental market. Market analysts should therefore evaluate Melbourne Airbnb activity as part of the broader accommodation and housing supply discussion.

8.2 Availability by Room Type

Figure: reports/figures/availability_by_room_type.png

Business interpretation:

Availability differs by room type, with shared rooms and hotel rooms showing higher average annual availability than entire homes and private rooms. Higher availability may indicate weaker demand, more flexible supply, or listings that are not actively booked throughout the year.

8.3 Top Neighbourhoods by Listing Count

Figure: reports/figures/top_neighbourhoods_by_listing_count.png

Business interpretation:

The Melbourne neighbourhood has the highest listing concentration. This indicates that Airbnb supply is spatially concentrated rather than evenly distributed across the city. Localised monitoring is therefore more useful than relying only on city-wide averages.

8.4 Review Score Distribution

Figure: reports/figures/review_score_distribution.png

Business interpretation:

Review score distribution helps assess customer satisfaction patterns. If scores are concentrated near the top end, small differences in rating may still matter commercially because many listings are competing within a narrow high-score band.

8.5 Reviews by Room Type

Figure: reports/figures/reviews_by_room_type.png

Business interpretation:

Entire homes/apartments have a higher average number of reviews than private rooms. This may indicate stronger guest activity, longer operating history, or greater visibility for entire-home listings.

8.6 Monthly Availability

Figure: reports/figures/monthly_availability_rate.png

Business interpretation:

Monthly availability patterns help identify potential seasonal or operational changes. Lower availability may suggest stronger demand or more host-blocked dates, while higher availability may indicate weaker demand or excess supply.

8.7 Host Concentration

Figure: reports/figures/top_hosts_by_listing_count.png

Business interpretation:

The host concentration chart identifies high-volume operators. This is useful for understanding whether supply is driven mainly by many casual hosts or by a smaller group of professional operators.

9. Statistical Findings

Statistical outputs were generated at:

```
reports/statistical_outputs/statistical_tests_summary.csv
```

Non-parametric tests were used because Airbnb marketplace variables such as availability, reviews, and review scores are unlikely to follow ideal normal distributions.

H1: Entire-home listings have different annual availability than private rooms

Test used: Mann-Whitney U test

* Sample size group 1: 17,703

* Sample size group 2: 6,563

* p-value: 0.000000

* effect size: 0.0659

Interpretation:

The result suggests that entire-home listings and private rooms have statistically different annual availability distributions. The effect size is small, so the practical difference should be interpreted carefully. However, it still indicates that accommodation type is relevant when analysing supply strategy.

H2: Superhost listings have different review scores than non-superhost listings

Test used: Mann-Whitney U test

* Sample size group 1: 6,781

* Sample size group 2: 12,735

* p-value: 0.000000

* effect size: 0.2590

Interpretation:

The result suggests that superhost status is associated with different review score distributions. The effect size is larger than the other tests in this project, indicating that superhost status may be a meaningful quality signal.

H3: Annual availability differs across neighbourhoods

Test used: Kruskal-Wallis test

* Total sample size: 24,491
* Neighbourhood groups: 30
* p-value: 0.000000
* effect size: 0.0565

Interpretation:

The result suggests that availability patterns differ across Melbourne neighbourhoods. The effect size is small, but the result supports neighbourhood-level monitoring rather than relying only on city-level averages.

H4: Weekend availability differs from weekday availability

Test used: Mann-Whitney U test

* Weekend records: 2,547,064
* Weekday records: 6,392,151
* p-value: 0.000000
* effect size: -0.0101

Interpretation:

The result suggests a statistically detectable difference between weekend and weekday availability. However, the effect size is very small, so the practical business impact may be limited. This is an example where statistical significance does not necessarily mean strong practical significance.

Data Science & ML Experiment: Listing Segmentation

To extend the analysis beyond descriptive reporting, I implemented a K-Means clustering experiment to segment Melbourne Airbnb listings into behavioural supply groups. The model used listing-level features including annual availability, estimated occupancy rate, review count, reviews per month, review score, host tenure, host listing count, room type, and superhost status.

The model produced four listing segments with a silhouette score of 0.3168. This indicates moderate separation between clusters, which is reasonable for real-world marketplace data where listing behaviour often overlaps rather than forming perfectly distinct groups.

The four listing segments were:

Segment	Listing Count	Segment Name	Business Interpretation
0	12,332	Low-availability active supply	Listings with low annual availability and high occupancy proxy, suggesting stronger utilisation.
1	8,483	High-availability casual or idle supply	Listings with high annual availability and lower occupancy proxy, suggesting casual, seasonal, or under-utilised supply.
2	2,804	Established high-review listings	Listings with very high review counts and strong review scores, suggesting established demand signals.
3	872	Low-rating low-activity listings	Listings with lower review scores, low review counts, and relatively high availability, suggesting potential quality or optimisation issues.

This segmentation creates a more practical market view than analysing listings only by room type or neighbourhood. For example, platform teams could prioritise Segment 3 for quality improvement support, while revenue or market teams could study Segment 2 to understand characteristics of established high-performing listings.

The experiment was intentionally framed as unsupervised segmentation rather than price prediction because the Melbourne dataset did not contain usable price values. This avoided creating unsupported predictive targets and kept the ML work aligned with validated data.

AI/ML Experiment: Review NLP Sentiment and Theme Analysis

To explore the unstructured review text, I implemented a lightweight NLP experiment using guest review comments. The analysis used text cleaning, lexicon-based sentiment scoring, and TF-IDF keyword extraction. This approach was selected because it is transparent, reproducible, and does not require external API keys or paid AI services.

The sentiment analysis classified review language into positive, neutral, and negative categories:

Sentiment	Review Count	Review Share
Positive	809,852	89.6%
Neutral	83,959	9.3%
Negative	9,590	1.1%

The results show that Melbourne Airbnb reviews are strongly skewed toward positive language. This aligns with the generally high review score distribution observed elsewhere in the project, but it also highlights a limitation: review sentiment alone may not be enough to identify listing quality issues because public review behaviour is highly positive overall.

The top review terms included “great”, “stay”, “place”, “location”, “apartment”, “clean”, “host”, “comfortable”, “recommend”, and “great location”. These terms suggest that guests most frequently describe the stay experience through location, cleanliness, host interaction, comfort, and ease of stay.

From a business perspective, this NLP analysis can support host success and operations teams by identifying the themes guests mention most often. It also provides a foundation for future work such as topic modelling, review-quality classification, or listing improvement recommendations based on guest language.

This AI/ML experiment was kept deliberately simple and explainable. A more advanced version could use transformer-based sentiment models or topic modelling, but the current approach is easier to audit and reproduce within the assessment scope.

10. Business Recommendations

Recommendation 1: Monitor Entire-Home Supply Closely

Entire homes/apartments dominate the Melbourne Airbnb dataset. Stakeholders should monitor this category separately because it represents the largest share of supply and may behave differently from private rooms or shared rooms.

Recommendation 2: Use Neighbourhood-Level Market Monitoring

The Melbourne neighbourhood has the highest listing concentration, while other areas such as Darebin show strong occupancy-rate proxy patterns. Market analysis should therefore be performed at neighbourhood level rather than relying only on city-wide averages.

Recommendation 3: Treat Occupancy Estimates Carefully

Occupancy estimates should be used as directional indicators only. Calendar unavailability may include actual bookings, host-blocked dates, maintenance, or inactive dates.

Recommendation 4: Track High-Volume Hosts

The largest host, Flexistayz, manages 292 listings. Host concentration analysis can help identify professional operators and distinguish them from casual hosts.

Recommendation 5: Use Superhost Status as a Quality Signal

The statistical analysis suggests that superhost listings have different review score distributions from non-superhost listings. This supports using superhost status as one signal of listing quality, while still considering other review and operational metrics.

Recommendation 6: Avoid Unsupported Price Conclusions

Because the Melbourne dataset did not contain usable price values, price and revenue conclusions should not be made from this dataset. Future work should use a city or dataset snapshot with valid price data if pricing analysis is required.

11. Assumptions and Decision Log

The full assumptions and decision log are documented in:

```
reports/decision_log.md
```

Major decisions include:

* Selecting one city for depth.

* Using Python for pipeline development.

* Using DuckDB for local analytical modelling.
* Excluding raw data from Git.
* Including report outputs for reviewer evidence.
* Pivoting away from price analysis because the Melbourne price fields were missing.
* Using non-parametric statistical tests.

12. Limitations and Caveats

Important limitations include:

* The project analyses only Melbourne.
* The dataset is a public scraped snapshot and may contain missing or inconsistent values.
* Price fields were missing, so price and revenue analysis were not performed.
* Calendar unavailable days are only an occupancy proxy.
* Review frequency is only a proxy for demand.
* Statistical tests do not prove causation.
* Machine learning and AI experiments were not completed due to prioritisation.

13. Future Improvements

With more time, the project could be extended by:

* Adding another city with complete price data.
* Building a Streamlit dashboard.
* Adding Docker for reproducibility.
* Implementing dbt models and tests.
* Adding machine learning price prediction using a dataset with valid price fields.
* Running NLP sentiment analysis on review text.
* Deploying the pipeline to a cloud platform.
* Adding automated data quality tests.
* Building a geographic map using neighbourhoods.geojson.

14. Reflection

This project was scoped to prioritise core data engineering and analytical quality.

The main trade-off was choosing depth over breadth. Instead of attempting every optional section, the project focused on building a reproducible pipeline, clean outputs, SQL analysis, EDA, and statistical testing.

A major learning was that real-world public datasets may not support the original analytical plan. In this case, missing price data required a responsible pivot from price analysis to supply, availability, reviews, host concentration, and quality signals.

The most important professional decision was to avoid inventing or imputing unsupported price values. This improves trust in the final analysis.

Appendix A. AI Usage Disclosure

The AI usage disclosure is documented in:

```
reports/ai_usage_disclosure.md
```

AI assistance was used for project planning, code structure guidance, debugging support, README drafting, and report structure guidance. All outputs were reviewed, executed, and validated locally.

Appendix B. Figure Gallery

Figure B.1: Availability By Room Type

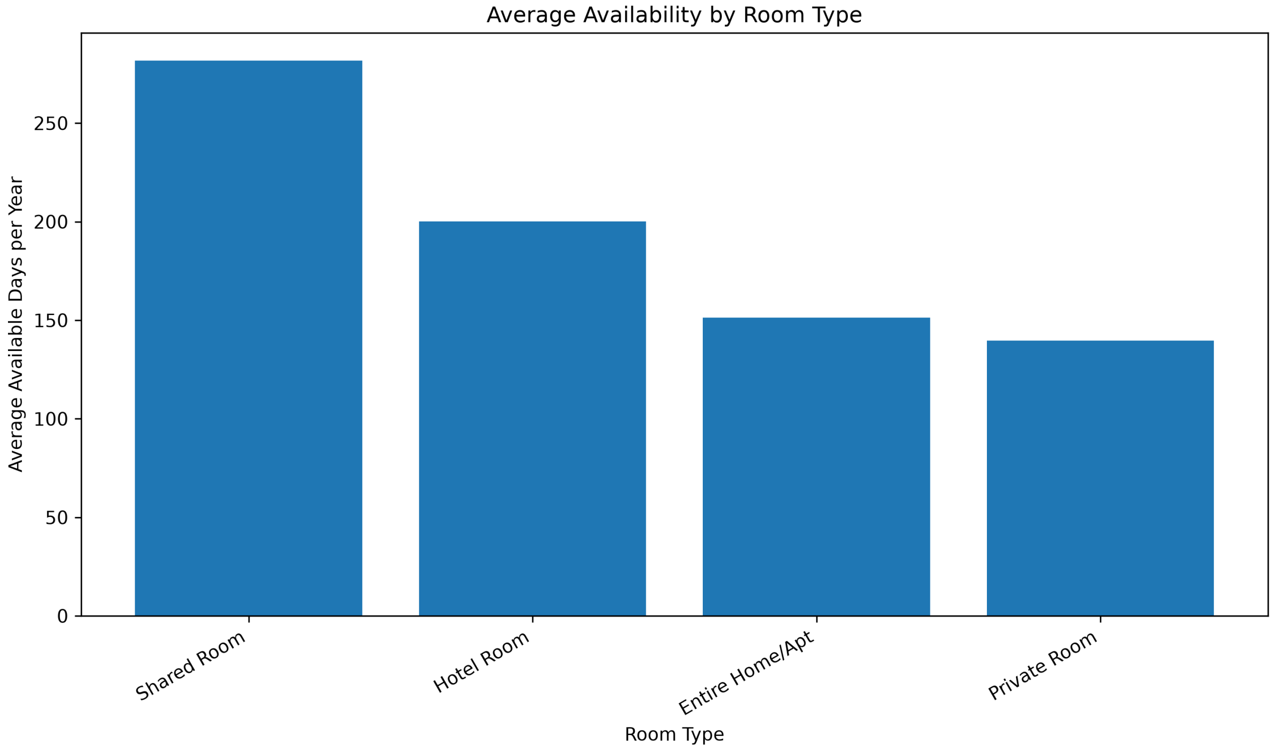


Figure B.2: Listing Segments Pca

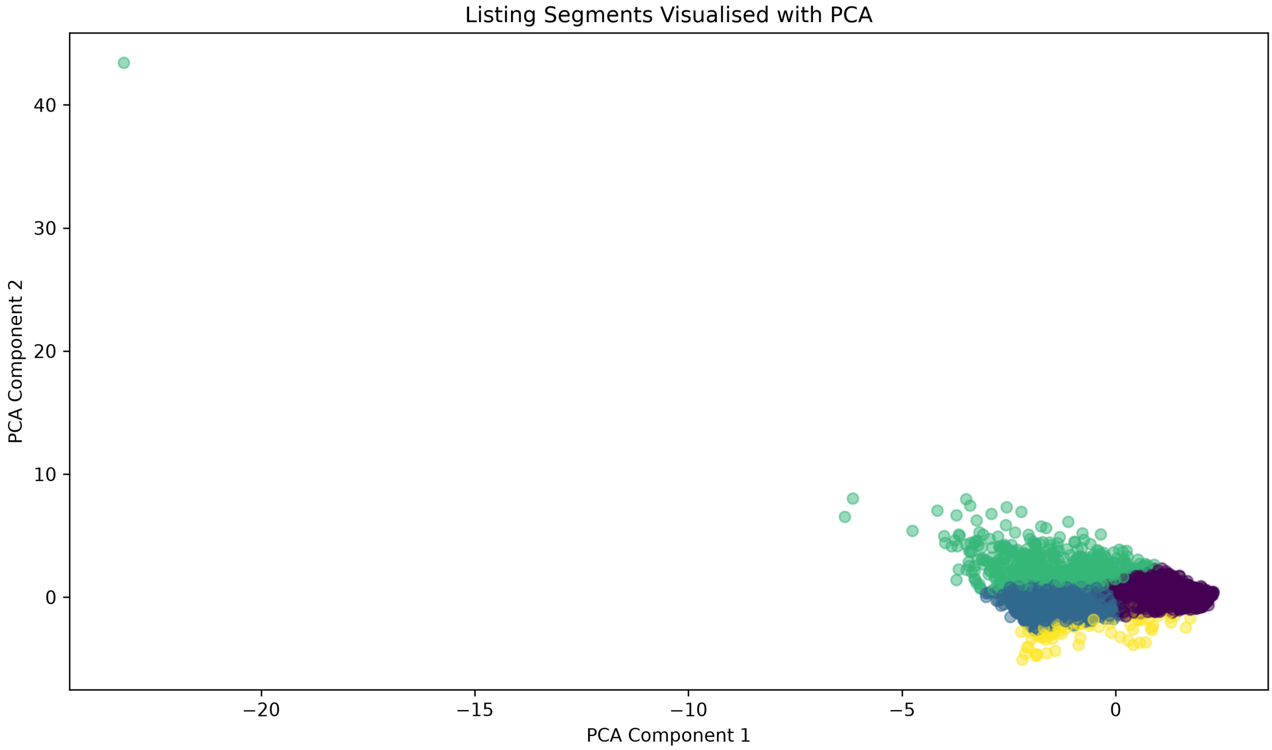


Figure B.3: Listing Supply By Room Type

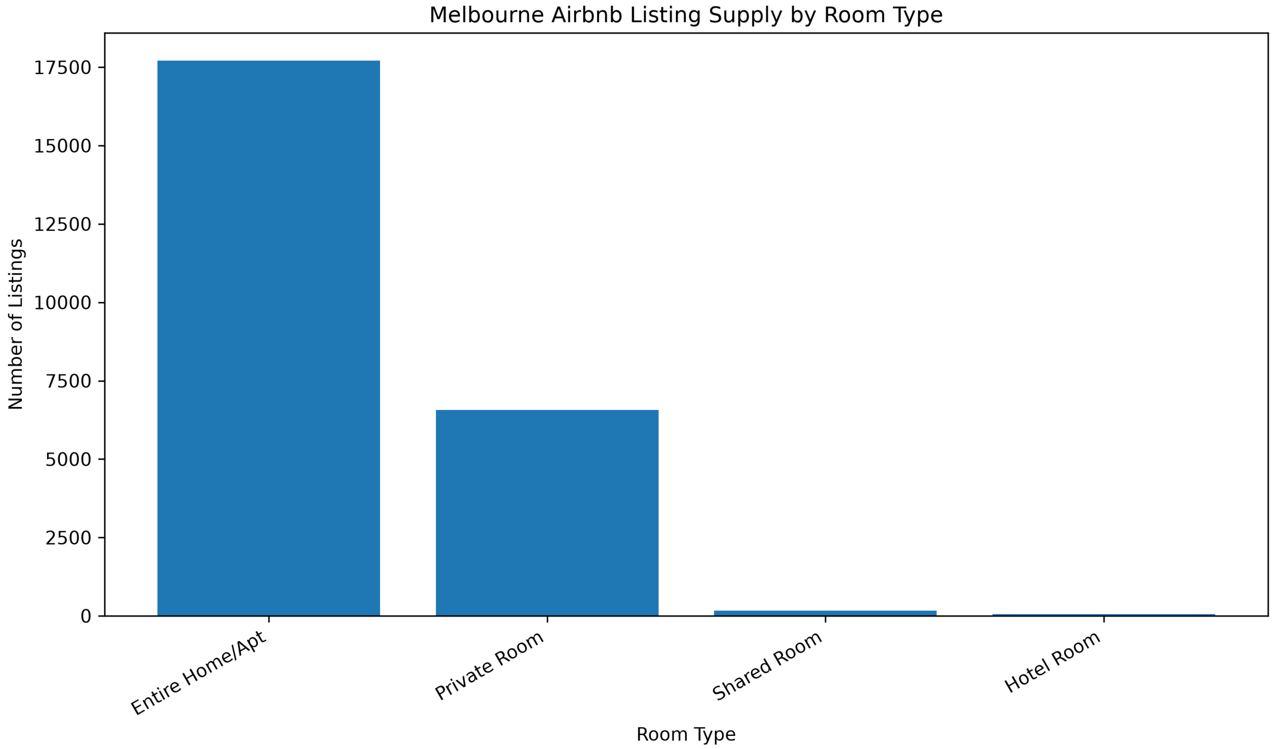


Figure B.4: Monthly Availability Rate

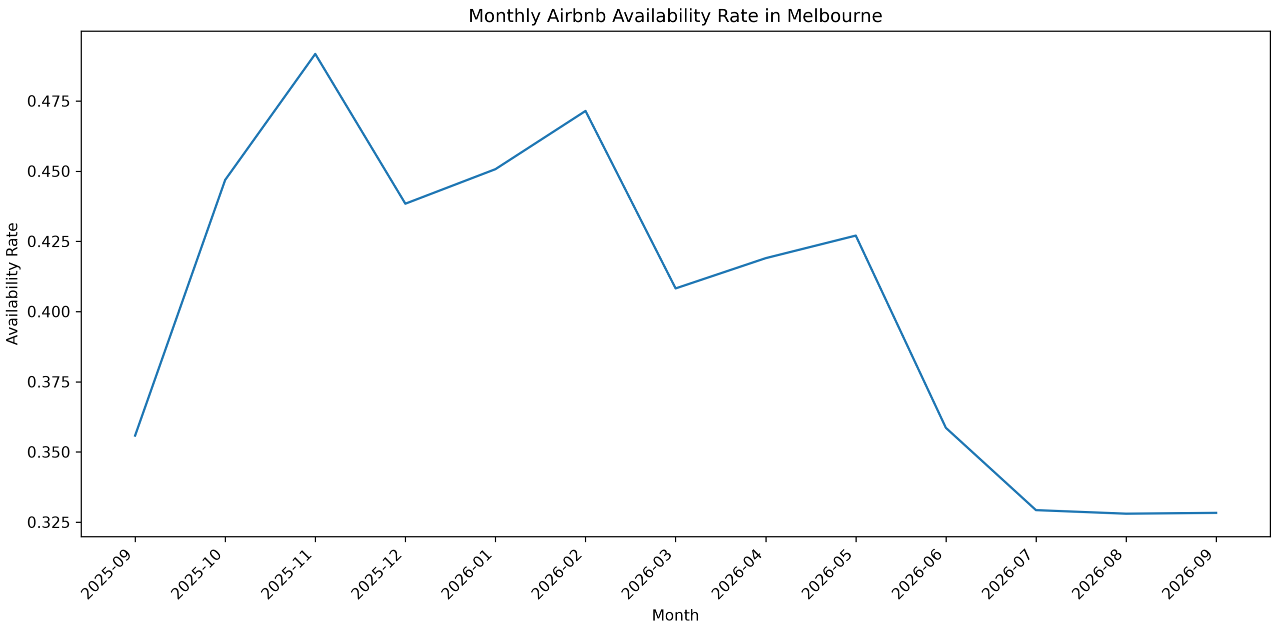


Figure B.5: Review Score Distribution

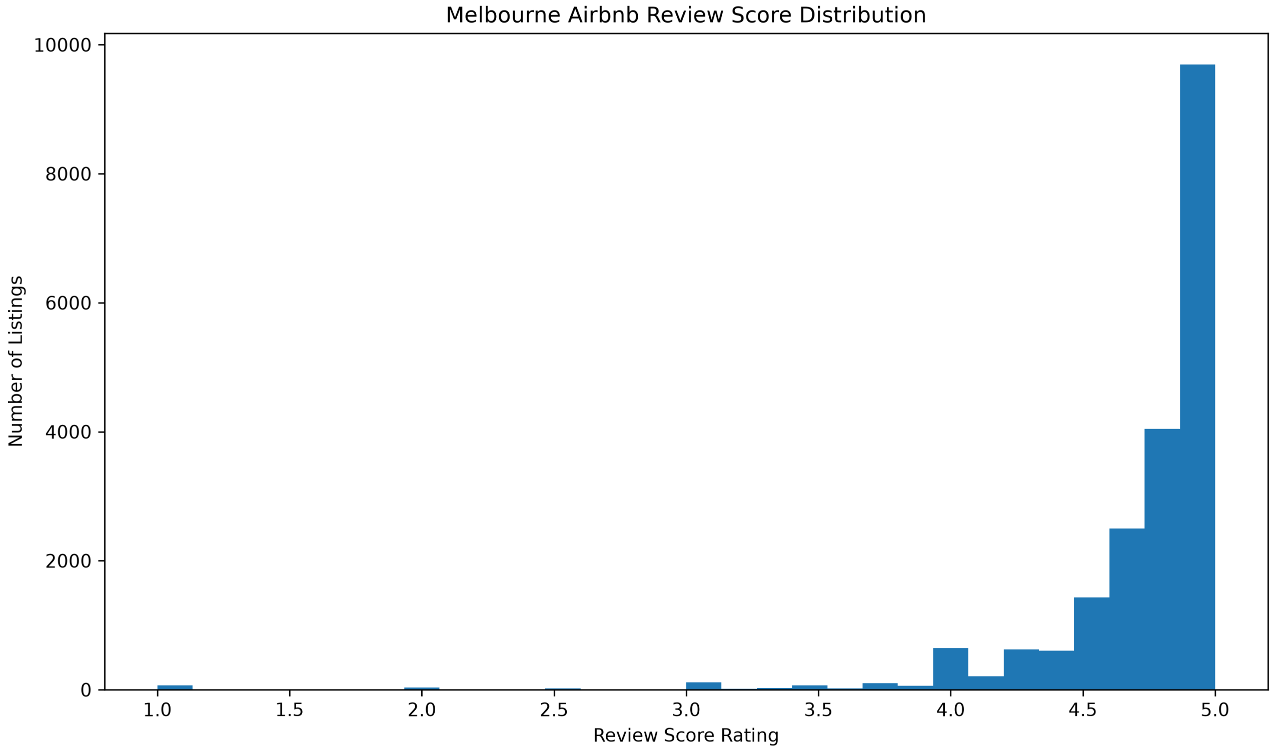


Figure B.6: Review Sentiment Distribution

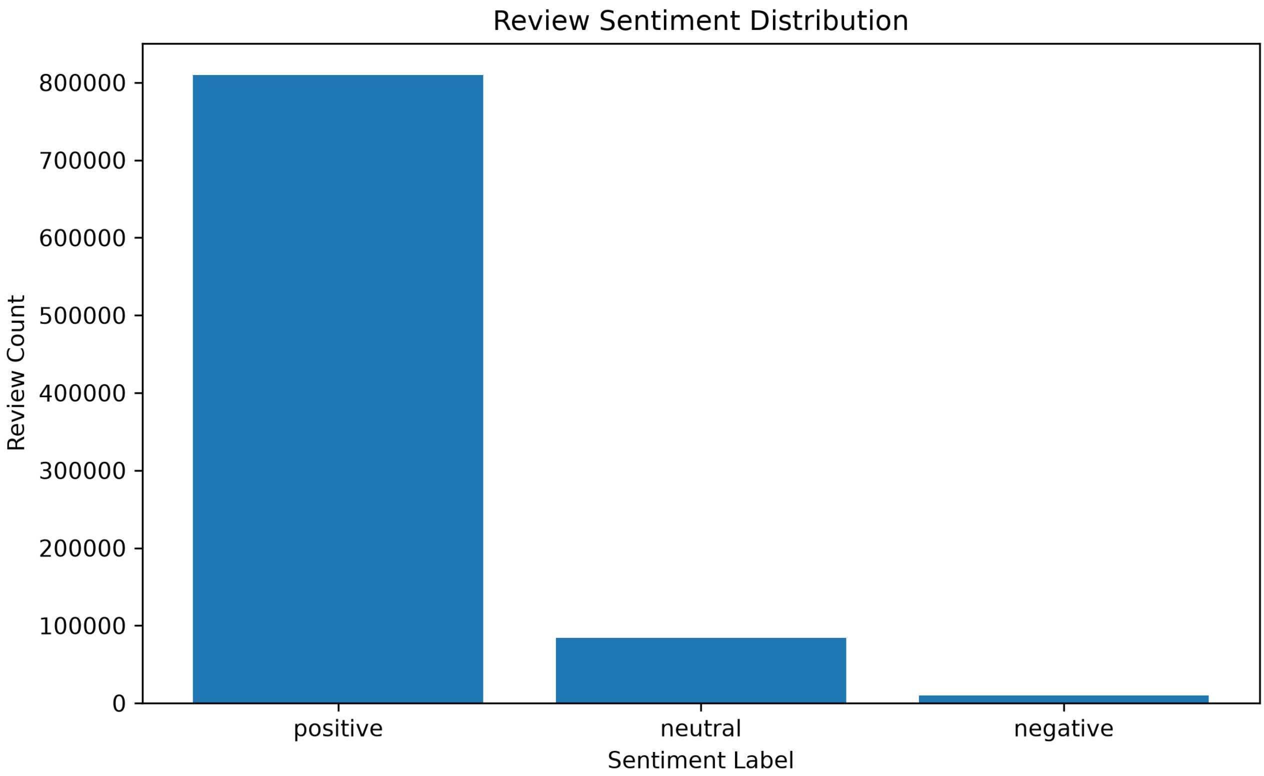


Figure B.7: Reviews By Room Type

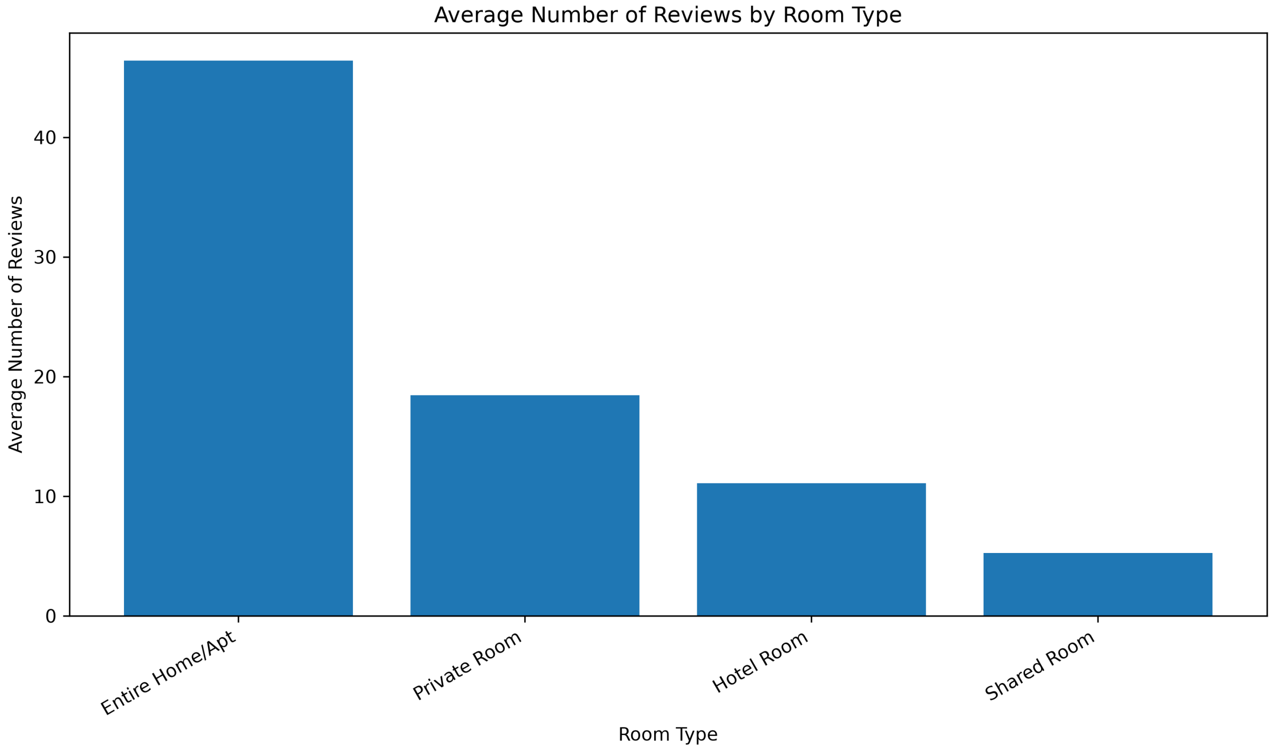


Figure B.8: Segment Profile Availability

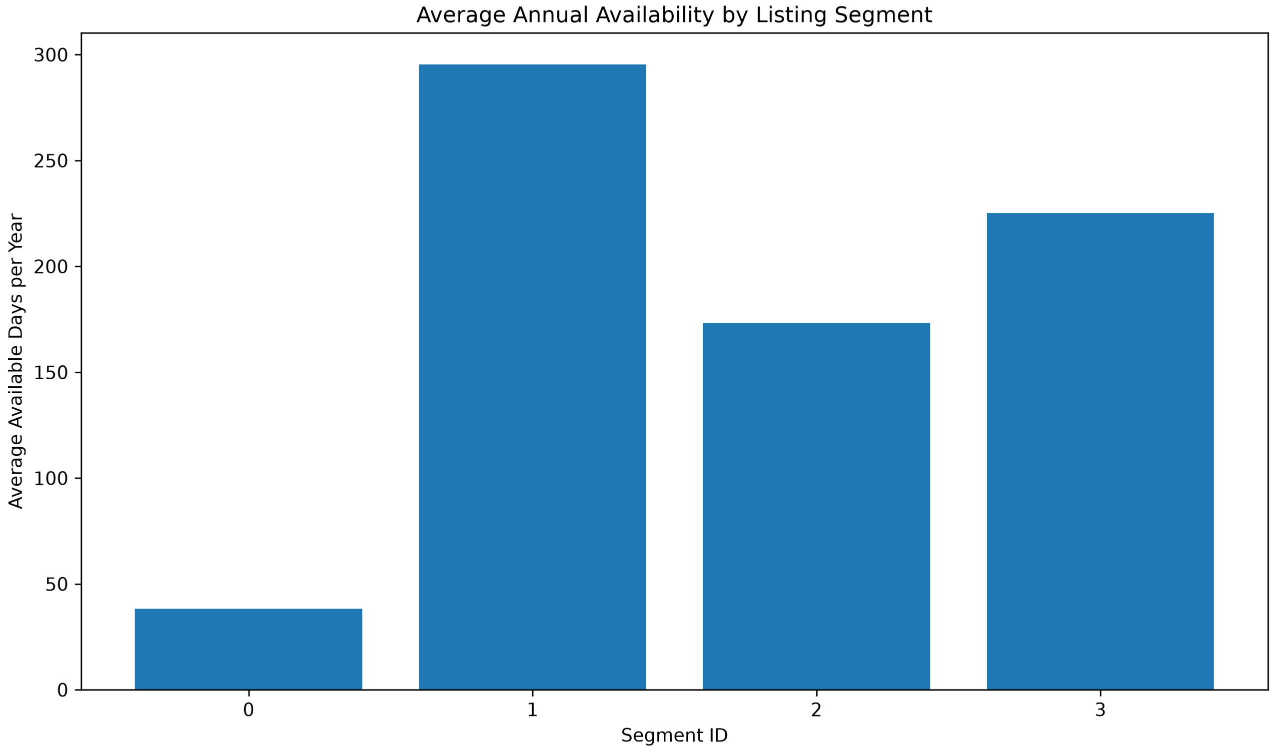


Figure B.9: Top Hosts By Listing Count

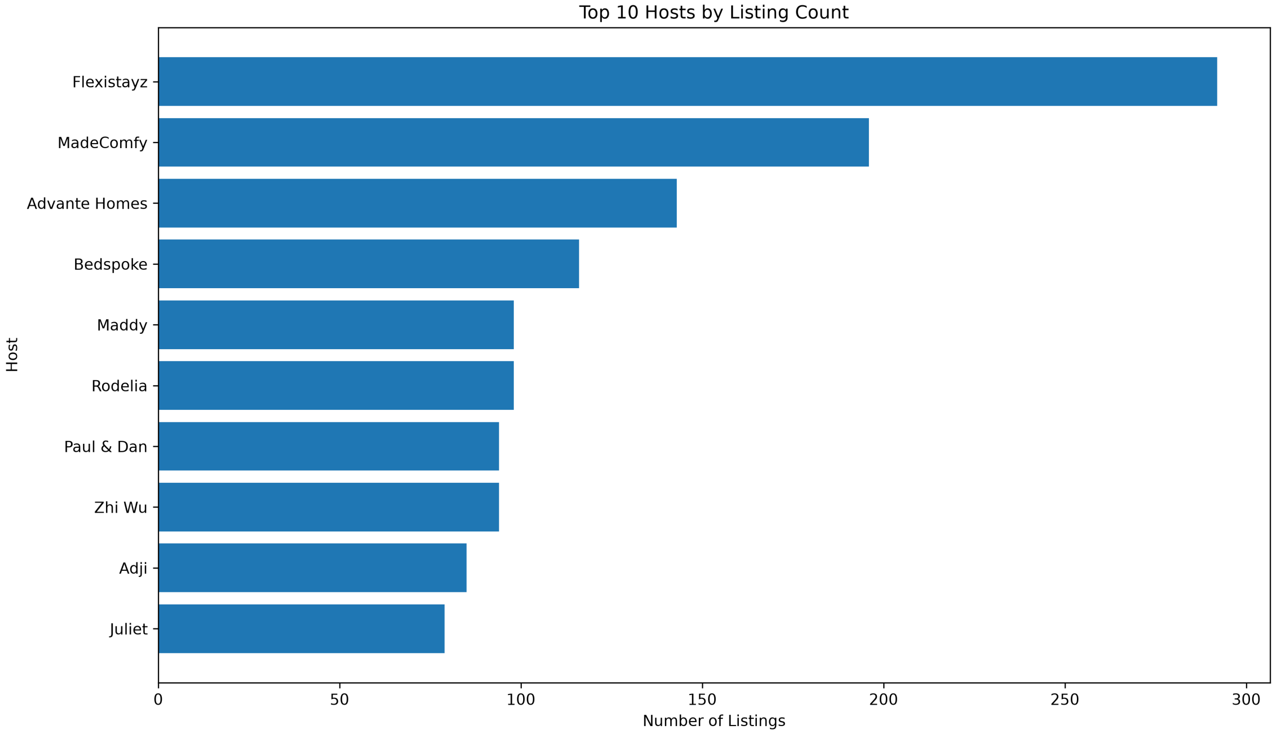


Figure B.10: Top Neighbourhoods By Listing Count

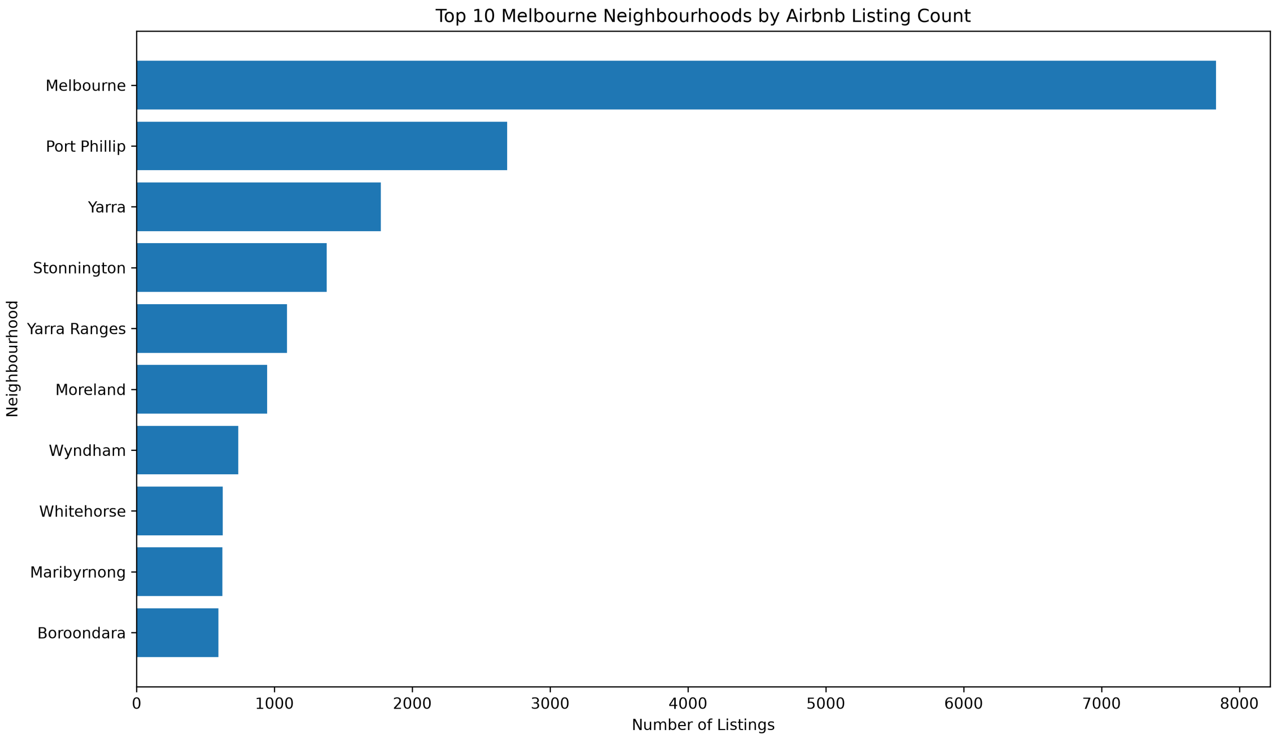
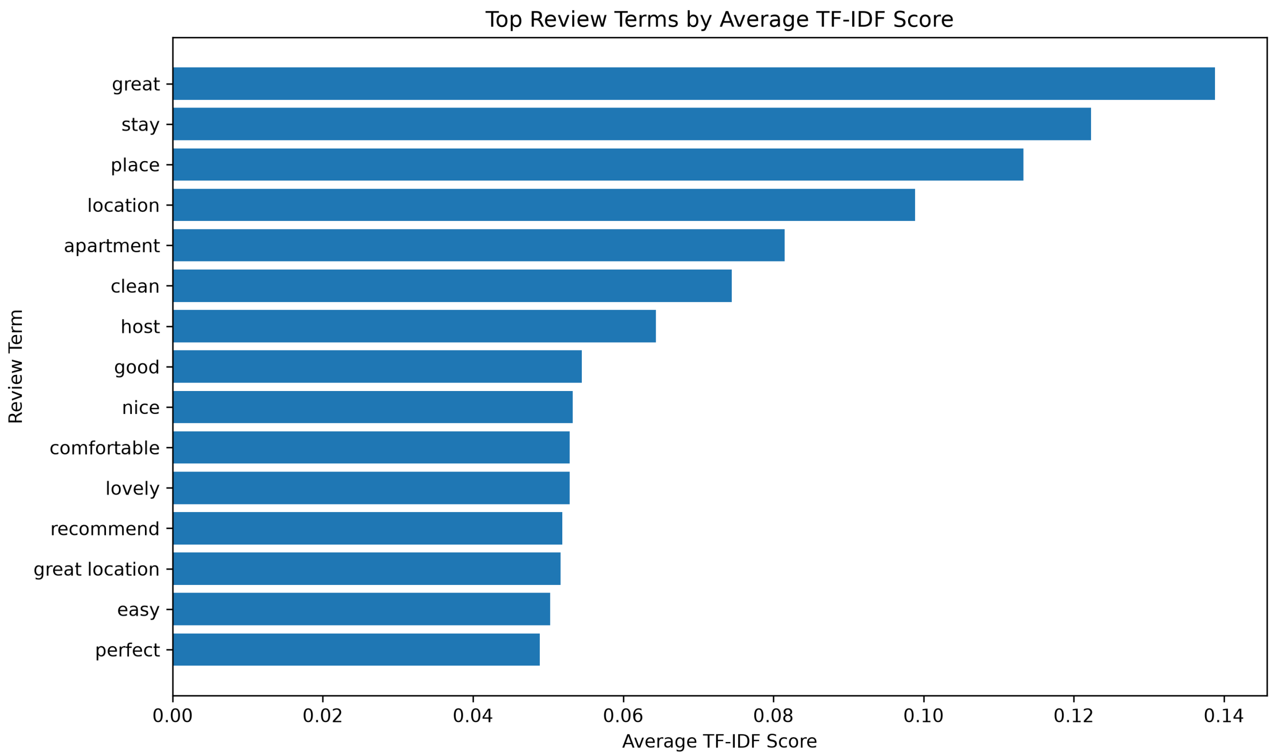


Figure B.11: Top Review Terms



Appendix C. SQL Output Tables

C.1: Query 01 Output

Source file: reports\sql_outputs\query_01_output.csv

room_type	listing_count	average_availability_365	median_availability_365	average_number_of_reviews
Entire Home/Apt	17703	151.37	130.0	46.39
Private Room	6563	139.74	89.0	18.45
Shared Room	170	281.64	364.0	5.25
Hotel Room	55	200.16	255.0	11.07

C.2: Query 02 Output

Source file: reports\sql_outputs\query_02_output.csv

neighbourhood_cleansed	listing_count	average_availability_365	average_number_of_reviews	average_review_score
Melbourne	7832	140.12	49.14	4.67
Port Phillip	2687	127.79	33.57	4.79
Yarra	1771	112.48	44.83	4.81
Stonnington	1380	123.04	31.37	4.78
Yarra Ranges	1092	226.94	82.29	4.85
Moreland	946	114.89	29.79	4.76
Wyndham	737	203.67	18.0	4.72
Whitehorse	624	158.19	18.76	4.67

C.3: Query 03 Output

Source file: reports\sql_outputs\query_03_output.csv

neighbourhood_cleansed	listing_count	average_occupancy_rate_estimate	average_availability_365	average_number_of_reviews
Darebin	584	0.7066	106.91	27.6
Yarra	1771	0.6918	112.48	44.83
Moreland	946	0.6836	114.89	29.79
Stonnington	1380	0.6623	123.04	31.37
Port Phillip	2687	0.6499	127.79	33.57
Boroondara	593	0.6314	134.53	24.26
Melbourne	7832	0.616	140.12	49.14
Glen Eira	527	0.6018	145.33	22.52

C.4: Query 04 Output

Source file: reports\sql_outputs\query_04_output.csv

host_is_superhost	listing_count	average_review_score	median_review_score	average_number_of_reviews
True	6781	4.87	4.9	78.59
	793	4.72	4.8	38.6
False	12735	4.66	4.79	29.86

C.5: Query 05 Output

Source file: reports\sql_outputs\query_05_output.csv

is_weekend	calendar_records	availability_rate
True	2547064	0.4019
False	6392151	0.412

C.6: Query 06 Output

Source file: reports\sql_outputs\query_06_output.csv

host_id	host_name	listing_count	average_availability_365	average_number_of_reviews
90729398	Flexistayz	292	94.66	5.8
279001183	MadeComfy	196	238.79	17.18
343442154	Advante Homes	143	93.47	22.69
446080599	Bedspoke	116	324.56	38.03
392306676	Rodelia	98	333.86	4.08
38814953	Maddy	98	328.31	5.2
244502930	Zhi Wu	94	278.09	11.67
1739996	Paul & Dan	94	237.94	191.37

C.7: Query 07 Output

Source file: reports\sql_outputs\query_07_output.csv

room_type	listing_count	average_number_of_reviews	average_reviews_per_month	average_availability_365	average_review_score
Entire Home/Apt	17703	46.39	1.48	151.37	4.74
Private Room	6563	18.45	0.73	139.74	4.72
Shared Room	170	5.25	0.53	281.64	4.46
Hotel Room	55	11.07	0.17	200.16	4.51

Appendix D. EDA Output Tables

D.1: Availability By Room Type

Source file: reports\eda_outputs\availability_by_room_type.csv

room_type	listing_count	average_availability_365	median_availability_365
Shared Room	170	281.63529411764705	364.0
Hotel Room	55	200.1636363636364	255.0
Entire Home/Apt	17703	151.36587019149297	130.0
Private Room	6563	139.73548682005182	89.0

D.2: Listing Supply By Room Type

Source file: reports\eda_outputs\listing_supply_by_room_type.csv

room_type	listing_count	average_availability_365	median_availability_365	average_number_of_reviews
Entire Home/Apt	17703	151.36587019149297	130.0	46.386205727842736
Private Room	6563	139.73548682005182	89.0	18.446899283864088
Shared Room	170	281.63529411764705	364.0	5.252941176470588
Hotel Room	55	200.1636363636364	255.0	11.072727272727272

D.3: Monthly Availability Rate

Source file: reports\eda_outputs\monthly_availability_rate.csv

month	calendar_records	availability_rate
2025-09	465329	0.35580847099579
2025-10	759221	0.4468712008756343
2025-11	734730	0.4917411838362391
2025-12	759221	0.4384046279014937
2026-01	759221	0.4507027598024817
2026-02	685748	0.4714384876076926
2026-03	759221	0.4082355466985238
2026-04	734730	0.4190083432009037

D.4: Reviews By Room Type

Source file: reports\eda_outputs\reviews_by_room_type.csv

room_type	listing_count	average_number_of_reviews	median_number_of_reviews	average_reviews_per_month
Entire Home/Apt	17703	46.386205727842736	15.0	1.4842277310384688
Private Room	6563	18.446899283864088	3.0	0.7257301656495205
Hotel Room	55	11.072727272727272	5.0	0.1658333333333333
Shared Room	170	5.252941176470588	1.0	0.5283333333333333

D.5: Top Hosts By Listing Count

Source file: reports\eda_outputs\top_hosts_by_listing_count.csv

host_id	host_name	listing_count	average_availability_365	average_number_of_reviews
90729398	Flexistayz	292	94.66438356164385	5.804794520547945
279001183	MadeComfy	196	238.78571428571428	17.183673469387756
343442154	Advante Homes	143	93.46853146853148	22.685314685314687
446080599	Bedspoke	116	324.5603448275862	38.03448275862069
38814953	Maddy	98	328.3061224489796	5.204081632653061
392306676	Rodelia	98	333.85714285714283	4.081632653061225
1739996	Paul & Dan	94	237.93617021276597	191.37234042553192
244502930	Zhi Wu	94	278.0851063829787	11.670212765957446

D.6: Top Neighbourhoods By Listing Count

Source file: reports\eda_outputs\top_neighbourhoods_by_listing_count.csv

neighbourhood_cleansed	listing_count	average_availability_365	average_number_of_reviews	average_review_score
Melbourne	7832	140.11963738508683	49.13981103166496	4.665213864306785
Port Phillip	2687	127.79493859322665	33.569036099739485	4.790529026217229
Yarra	1771	112.48277809147376	44.82552230378317	4.811338073691015
Stonnington	1380	123.04057971014493	31.370289855072464	4.780182815356491
Yarra Ranges	1092	226.9386446886447	82.29212454212454	4.848567222767419
Moreland	946	114.89323467230444	29.79069767441861	4.755273885350318
Wyndham	737	203.6729986431479	18.00135685210312	4.715637342908439
Whitehorse	624	158.19070512820514	18.75961538461583	4.665352941176471

Appendix E. Statistical Output Tables

E.1: Statistical Tests Summary

Source file: reports\statistical_outputs\statistical_tests_summary.csv

hypothesis	test_used	status	null_hypothesis	alternative_hypothesis	sample_size_group_1	sample_size_group_2	test_statistic	p_value	effect_size	business_interpretation
H1: Entire-home listings have different annual availability than private rooms	Mann-Whitney U test	Completed	There is no annual availability distribution difference between entire-home listings and private rooms.	There is an annual availability distribution difference between entire-home listings and private rooms.	17703	6563	61918973.0	0.0	0.0659	This tests whether different accommodation types behave differently in terms of yearly availability. A significant result suggests supply strategy differs by room type.
H2: Superhost listings have different review scores than non-superhost listings	Mann-Whitney U test	Completed	There is no review score distribution difference between superhost and non-superhost listings.	There is a review score distribution difference between superhost and non-superhost listings.	6781	12735	54361346.0	0.0	0.259	This tests whether superhost status is associated with guest satisfaction. A significant result suggests superhost status may be a useful quality signal.
H3: Annual availability differs across neighbourhoods	Kruskal-Wallis test	Completed	Annual availability distributions are the same across neighbourhoods.	At least one neighbourhood has a different annual availability distribution.	24491	30	1411.8122	0.0	0.0565	This tests whether supply availability patterns differ by location. A significant result suggests neighbourhood-level market monitoring is needed.
H4: Weekend availability differs from weekday availability	Mann-Whitney U test	Completed	Weekend and weekday availability distributions are the same.	Weekend and weekday availability distributions are different.	2547064	6392151	8058345464139.0	0.0	-0.0101	This tests whether availability patterns differ between weekends and weekdays. A significant result may indicate seasonal or demand-driven booking behaviour.